Shamar Lindo

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EDUCATION

The University of North Carolina at Charlotte **December 2028**
Bachelor of Science in Mechanical Engineering

- Relevant Coursework: Statics, Mechanics of Materials, Thermodynamics, Dynamic Systems, Computational Methods

SKILLS

- **Design & CAD:**SolidWorks, SOLIDWORKS Plastics, PTC Creo Parametric, 3D Part Modeling, Assembly Design, GD&T, Technical Drawings, DFM, Reverse Engineering, Mold Flow Analysis
- **Manufacturing & Materials:** Injection Molding, Twin-Screw Extrusion, ASTM Testing, Material Characterization, FTIR, TGA, SEM, Optical Microscopy, Raw Material Qualification, Manual Vertical Mill, Engine Lathe
- **Programming & Electronics:** Python, MATLAB, C/C++, pandas, NumPy, Matplotlib, Processing, HTML/CSS, Data Analysis, Data Visualization
- **Software:** Microsoft Excel, Microsoft Word
- **Languages:** English, Jamaican Patois

WORK EXPERIENCE

Undergraduate Research Assistant **May 2026 - July 2026**
The University of North Carolina at Charlotte *Charlotte, NC*

- **Raw Material Qualification:** Sourced and characterized hemp hurd from regional suppliers via moisture, ash, and optical microscopy analysis, supporting supplier selection targeting a 71 percent cost reduction.
- **Cellulose Extraction:** Supported chemical extraction trials (alkali treatment, bleaching, acid hydrolysis) to isolate high-purity cellulose powder, assisting with sample prep, FTIR/TGA/SEM characterization, and batch production.

Footwear Sales Associate **June 2025 - Present**
Dick's Sporting Goods *Charlotte, NC*

- Deliver an exceptional, inclusive shopping experience by greeting customers, assessing their needs, and applying strong footwear product knowledge to drive sales.
- Maintain company standards through merchandising, signage, and inventory presentation while promoting loyalty, warranty, and credit card programs.

Bagger **June 2023 - July 2024**
Harris Teeter *Charlotte, NC*

- Delivered fast, accurate front-end service in a high-volume grocery environment, bagging orders, providing carryout, and loading customer vehicles to uphold the “Take Excellent Care of Our Customers” standard.
- Kept checkout lines moving during peak-hour rushes by coordinating with cashiers and communicating clearly with customers, while handling sustained physical demands including 40-pound lifts.

PROJECTS

Injection-Molded Part Redesign | SolidWorks, DFM, Mold Flow **September 2026 - Oct 2026**

- Reverse-engineering a molded consumer product housing from physical teardown and caliper measurement, building a validated SolidWorks model with surface-lofted contours, shelled thin walls, and draft applied to a non-planar parting line.
- Conducting design-for-manufacturing analysis using Draft Analysis, DFMXpress, and SOLIDWORKS Plastics fill simulation, correlating predicted sink marks and weld lines to defects observed on the physical part.
- Redesigning the housing to DFM standards — uniform wall thickness, corrected rib and boss ratios, eliminated undercuts — and producing a fully dimensioned GD&T drawing package.

Mold Flow Simulation Analyzer | Python, pandas, NumPy, Matplotlib **September 2026 - Oct 2026**

- Building a Python tool that parses SOLIDWORKS Plastics simulation output across a parametric sweep of wall thickness and gate positions, extracting fill time, injection pressure, and cooling time for each configuration.
- Automating trend plotting and optimization to identify the wall thickness minimizing cycle time subject to injection pressure limits, with validation against analytically generated test cases.

Toiletries Case Assembly | PTC Creo, HTML/CSS, Technical Writing **August 2025 - September 2025**

- Collaborated with a cross-functional team to design a compact, watertight travel toiletries case, modeling the full assembly in PTC Creo Parametric with precise tolerance fits for a reliable seal.
- Authored a comprehensive user manual and developed a responsive product advertisement website in HTML/CSS to market the product’s capabilities.

Ultrasonic Radar System | Arduino (C/C++), Processing, HC-SR04 **July 2026 - August 2026**

- Built a functional radar system using an Arduino Uno, HC-SR04 ultrasonic sensor, and SG90 servo motor to detect and track objects within sensor range.
- Programmed sensor sweep and distance-measurement logic in the Arduino IDE and developed a real-time radar visualization interface in Processing.